# STUDENT ASSIGNMENT AND TASK TRACKING SYSTEM

## A PROJECT REPORT

*Submitted by*

**A VISHAL**

(2116241501247)

*in partial fulfilment of the End Semester Examination for the course*

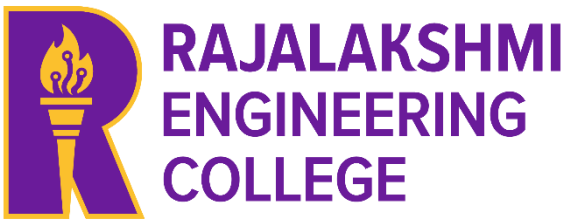
**CB23F35 –**

**LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR**

*in*

**BACHELOR OF TECHNOLOGY**

**Artificial Intelligence and Machine Learning**




**RAJALAKSHMI ENGINEERING COLLEGE**

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

**2026-2027**

**RAJALAKSHMI ENGINEERING COLLEGE**

(An Autonomous Institution, Affiliated to Anna University, Chennai)

# BONAFIDE CERTIFICATE

Certified that this project report titled "STUDENT ASSIGNMENT AND TASK TRACKING SYSTEM" is the bonafide work of A VISHAL (2116241501247), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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| INTERNAL EXAMINER | EXTERNAL EXAMINER |

# DECLARATION

I, A VISHAL (241501247), hereby declare that the project report titled "STUDENT ASSIGNMENT AND TASK TRACKING SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

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# ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman **Thiru S. Meganathan,** Chairperson **Dr. (Mrs.) Thangam Meganathan** and Vice-Chairman **Mr. Abhay Shankar Meganathan (B.E., M.S.)** for providing the infrastructure and facilities that made this work possible.

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

**A VISHAL**

# ABSTRACT

Managing academic assignments and deadlines is an important part of student life. Students often handle multiple subjects, each with different assignments, submission dates, and priorities. When assignment details are maintained in notebooks, spreadsheets, or separate applications, it can become difficult to track pending work, identify approaching deadlines, and maintain an organised record of academic tasks. A centralised assignment tracking system can help students manage these activities more effectively.

The Student Assignment and Task Tracking System is a low-code application developed using Zoho Creator. It maintains subject details and assignment records, including titles, descriptions, due dates, priorities and statuses. Forms provide structured data entry, while reports and a dashboard help users review assignments and task progress. Three Deluge workflows support due-date validation, automatic priority assignment for approaching deadlines and email notification when a student task is marked as completed.

The project demonstrates the application of low-code development to an academic task-management problem. The completed application provides an organised method for recording assignments, monitoring deadlines and reviewing completion progress. The implementation was reviewed using sample subject, assignment and student task records. Workflow execution and email delivery should be considered verified only where the corresponding test was performed.



**Keywords:** *Low-Code Development, Zoho Creator, Assignment Management, Task Tracking, Deadline Monitoring, Workflow Automation.*

# PROJECT AT A GLANCE

| Attribute | Details |
|---|---|
| Project Title | Student Assignment and Task Tracking System |
| Application Domain | Education / Academic Task Management |
| Student Name & Register No. | A VISHAL – 241501247 |
| Zoho Creator Edition | Free edition |
| Application Name (Link Name) | Student Assignment and Task Tracking System |
| Published / Live URL | https://creatorapp.zoho.in/vishal.a.2024.aiml_rajala284/student-assignment-tracking-system#Page:DashBoard |
| Demo Credentials (per role) | Administrator: authorised Zoho account; Student: not configured |
| Number of Forms / Subforms | 3 forms / 0 subforms |
| Number of Reports (by type) | 3 main list reports: All Assignments, All Subjects and All Student Tasks; dashboard includes summary views |
| Number of Pages / Dashboards | 1 dashboard page |
| Number of Workflows | 3 form workflows; no schedule, approval, blueprint or batch workflow |
| Deluge Functions / Custom Actions | 3 Deluge workflow scripts; no separate custom functions |
| Integrations | No external integrations |
| Roles & Profiles Defined | Administrator access; separate Faculty and Student profiles not configured |
| Environments Used | Development, Stage and Production (version 1.0) |
| Date of Production Deployment | 21 October 2026 |

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# LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| API | Application Programming Interface |
| CRUD | Create, Read, Update, Delete |
| ER | Entity–Relationship |
| JSON | JavaScript Object Notation |
| LCAP | Low-Code Application Platform |
| OAuth | Open Authorization |
| RBAC | Role-Based Access Control |
| REST | Representational State Transfer |
| SDLC | Software Development Life Cycle |
| SSO | Single Sign-On |
| UAT | User Acceptance Testing |
| UI | User Interface |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

Academic task management is an important part of student life, as students handle assignments, projects and assessments across multiple subjects. Each subject may have different submission deadlines and requirements, making it necessary to organise tasks and monitor their completion.

Currently, students often track assignments using notebooks, spreadsheets, calendars or messaging applications. Managing information across these different sources can make it difficult to identify pending assignments, remember deadlines and prioritise tasks.

The Student Assignment and Task Tracking System is proposed as a low-code application developed using Zoho Creator. It aims to provide a centralised platform for maintaining subject details, recording assignments, tracking due dates and updating completion status. Reports and a dashboard can help students view their academic workload and monitor their progress efficiently.

## 1.2 Problem Statement

Students manage assignments from multiple subjects, each with different deadlines and priorities. Using notebooks, spreadsheets and messaging applications to track assignments can make it difficult to maintain organised records, monitor pending work and complete tasks on time. Therefore, a centralised Student Assignment and Task Tracking System is proposed to simplify assignment management and improve visibility of academic progress.

- Assignment details are scattered across different sources.
- Upcoming deadlines may be forgotten, leading to late submissions.
- Tracking pending and completed assignments manually is time-consuming.
- Prioritising assignments across multiple subjects is difficult.
- There is no single dashboard to monitor overall assignment progress.

## 1.3 Objectives

- To design forms and a data model for managing subjects, assignments, deadlines, priorities and completion status.
- To implement due-date validation and automatic priority assignment using Deluge scripting and workflows.

- To develop reports and a dashboard for monitoring assignment records and completion progress.
- To configure workflow automation for notifying students when tasks are marked as completed.
- To evaluate the application through functional review and identify limitations and possible future enhancements.

### 1.4 Scope of the Project

In-Scope:

- Managing subject details and assignment records, including due dates, priorities and completion status.
- Validating assignment details and identifying overdue tasks using Deluge scripts and workflows.
- Generating reports for pending, completed and overdue assignments.
- Providing a dashboard to monitor assignment progress across subjects.
- Defining administrator and student access permissions.
- Developing a web-based application using Zoho Creator, with mobile access where supported.

Out-of-Scope:

- Integration with college learning management systems or external academic portals.
- Automatic retrieval of assignments from emails or college websites.
- AI-based assignment recommendations or automatic prioritisation.
- Online assignment submission, grading and plagiarism detection.
- Attendance management, examination results and complete student information management.
- External SMS notifications or third-party calendar integrations.

### 1.5 Justification for a Low-Code Approach

Zoho Creator is selected for this project because it provides visual form building, report generation, dashboards and workflow automation, which help reduce development time and programming effort. Deluge scripting can be used to implement custom logic such as deadline validation, assignment status updates and overdue task identification. The

platform provides built-in cloud hosting, mobile access and user permission controls, reducing the need to manage separate infrastructure. Its visual development environment also makes the application easier to maintain and modify. Therefore, Zoho Creator is suitable for developing the Student Assignment and Task Tracking System in a cost-effective and efficient manner.

### 1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

# CHAPTER 2
# LITERATURE SURVEY AND SYSTEM STUDY

## 2.1 Study of the Existing System

In the existing approach, students receive assignment details and deadlines through classroom announcements, messaging applications, emails or learning management systems. Students record these details in notebooks, calendars or spreadsheets and manually update their progress as they complete each assignment. They must check different sources to remember deadlines and identify pending tasks.

The typical process is:

Receive Assignment → Record Details → Track Deadline → Complete Assignment → Update Status → Review Pending Tasks

This process relies on manual record-keeping and regular updates by students, making it difficult to maintain a consolidated view of assignments across multiple subjects.

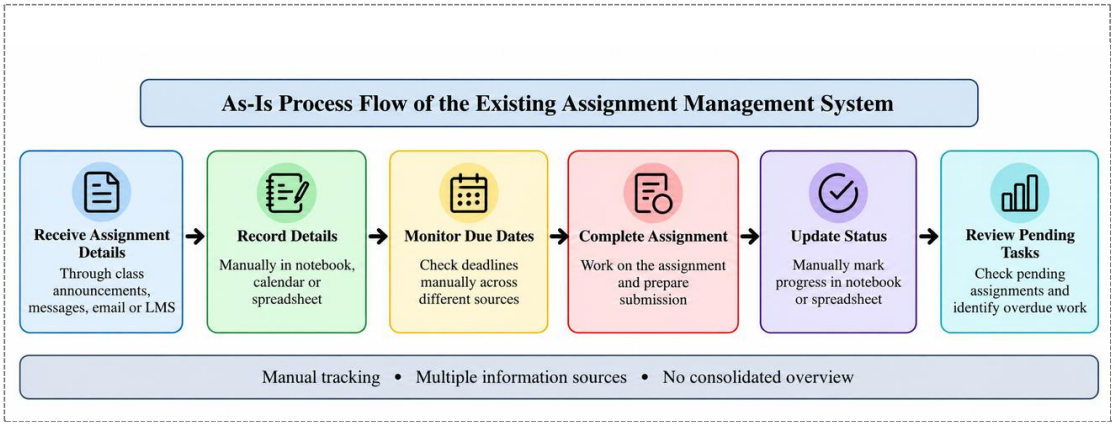



**Figure 2.1:** As-Is Process Flow of the Existing Assignment Management System

## 2.2 Limitations of the Existing System

- Assignment details are scattered across notebooks, spreadsheets, calendars and messaging applications, making information difficult to organise.
- Students may forget deadlines or overlook pending assignments due to manual tracking.
- Monitoring assignment status and prioritising tasks across multiple subjects is time-consuming without a centralised dashboard.

## 2.3 Low-Code versus Traditional Development

**Table 2.1:** Comparison of Low-Code and Traditional Development for the Proposed Application

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| Development time | 8–10 weeks | 2–3 weeks |

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| Skills required | Programming, database and web development skills | Basic application design and Deluge scripting skills |
| Hosting & infrastructure | Separate hosting and server configuration | Built-in cloud hosting |
| Mobile application | Separate mobile development or responsive design | Access through Zoho Creator mobile app |
| Security & access control | Custom authentication and permission implementation | Built-in user management and configurable permissions |
| Maintenance & change | Code changes and redeployment may be required | Visual configuration and quicker modifications |
| Customisation limits | High flexibility and control | Limited by platform capabilities |
| Cost | Higher development and maintenance costs | Lower development effort and infrastructure costs |

## 2.4 Literature Survey

**Table 2.2:** Summary of Literature Survey

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 1 | Gao et al. (2026) | What Does Current Research Say About the Viability of Low-Code Development? | Review of research identifies faster development as a benefit, while quality and complexity depend on context. | Supports using low-code for rapid application development. |
| 2 | Journal of Systems and Software (2025) | Adoption of Low-Code and No-Code Development: A Systematic Literature Review | Reviews adoption, benefits and challenges of low-code/no-code platforms. | Provides background for selecting a low-code approach. |
| 3 | Gartner (2024) | Critical Capabilities for Enterprise Low-Code Application Platforms | Examines platform capabilities such as application development, integration, governance and security. | Helps evaluate low-code platform suitability. |
| 4 | Zoho Corporation (n.d.) | Zoho Creator Help Documentation | Explains forms, reports, pages and application data management. | Guides the design of subject and assignment forms and reports. |

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 5 | Zoho Corporation (n.d.) | Understanding Workflows – Zoho Creator Help | Describes event-triggered automation, scheduled actions and reusable functions. | Supports planned Deluge scripting and assignment workflow automation. |

### 2.5 Emerging Trends in Low-Code Development

Low-code development is evolving through trends such as AI-assisted application building, citizen development, hyper-automation and composable applications. AI-assisted tools help developers design applications more quickly, while citizen development enables users with limited programming experience to build applications using visual interfaces. Hyper-automation combines workflows and scripting to reduce repetitive manual tasks, whereas composable applications allow reusable components to be combined to meet specific requirements. These trends influenced the design of the Student Assignment and Task Tracking System by encouraging the use of Zoho Creator's visual form builder, reports, dashboards and Deluge scripting. The proposed application uses these low-code features to organise assignment records, monitor deadlines and automate selected task-management activities with reduced development effort.

### 2.6 Proposed System

The Student Assignment and Task Tracking System was developed using Zoho Creator to provide a centralised platform for managing subject details and academic assignments. It uses Subject, Assignment and Student Task forms, reports and a dashboard. Deluge workflows validate due dates, set High priority for assignments due within the next three days, and send an email notification when a student task is marked Completed. The current implementation does not include external API integration or separately configured Faculty and Student permission profiles.

# CHAPTER 3
# REQUIREMENT ANALYSIS

## 3.1 Stakeholders and User Roles

**Table 3.1:** Stakeholders and User Roles

| Role | Description | Key Responsibilities in the Application |
|---|---|---|
| Administrator | Manages application configuration and user access. | Manage users, configure settings, view and manage all assignment data. |
| Faculty / Teacher | Creates assignments for students and monitors submissions or completion progress. | Create assignments, set due dates and priorities, update assignment details, and monitor completion status. |
| Student | Views assignments assigned by faculty and tracks academic tasks. | View assigned tasks, check deadlines, and update personal task progress or completion status. |

## 3.2 Functional Requirements

**Table 3.2:** Functional Requirements

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-01 | The system shall allow faculty to create a new assignment with mandatory fields such as title, subject, description and due date. | Faculty | High |
| FR-02 | The system shall allow faculty to set assignment priorities and update assignment details after creation. | Faculty | High |
| FR-03 | The system shall provide a Student Task record linked to an assignment so that task progress can be recorded. | Student | High |
| FR-04 | The system shall allow completion status to be updated in the Student Task form. | Student | High |
| FR-05 | The system shall display assignment summary information, priority distribution and completion status on the dashboard. | Faculty, Administrator | Medium |
| FR-06 | The system may identify overdue assignments in a future enhancement; automatic overdue identification is not implemented in the current version. | Faculty, Administrator | High |
| FR-07 | The system shall validate mandatory assignment fields before saving a record. | Faculty | High |
| FR-08 | The application shall use available Zoho Creator user-management settings; separate role-based profiles were not configured in the current version. | Administrator | Medium |

## 3.3 Non-Functional Requirements

**Table 3.3:** Non-Functional Requirements

| NFR ID | Category | Requirement |
|---|---|---|
| NFR-01 | Performance | Reports containing up to 1,000 records shall load within 5 seconds under normal operating conditions. |
| NFR-02 | Security | Users shall access only the data permitted by their assigned role. |
| NFR-03 | Usability | The application shall be usable through a web browser and the Zoho Creator mobile app. |
| NFR-04 | Availability | The application shall be accessible whenever the Zoho Creator service is available and the user has an internet connection. |
| NFR-05 | Maintainability | The application shall allow authorised users to modify forms, reports and workflows with minimal changes to the overall design. |
| NFR-06 | Auditability | Record changes shall be traceable through available audit-trail features where supported and configured. |

## 3.4 Hardware and Software Requirements

**Table 3.4:** Hardware and Software Requirements

| Category | Specification |
|---|---|
| Platform | Zoho Creator Education Edition |
| Scripting language | Deluge (Data Enriched Language for the Universal Grid Environment) |
| Browser | Google Chrome / Microsoft Edge / Mozilla Firefox (latest version) |
| API testing tool | Not used |
| Other Zoho services | None initially; Zoho Creator built-in forms, reports, pages and workflows |
| Mobile | Not implemented / not demonstrated |
| Client hardware | Intel Core i3/i5 or above, 8 GB RAM, internet connection |

## 3.5 Use Case Model

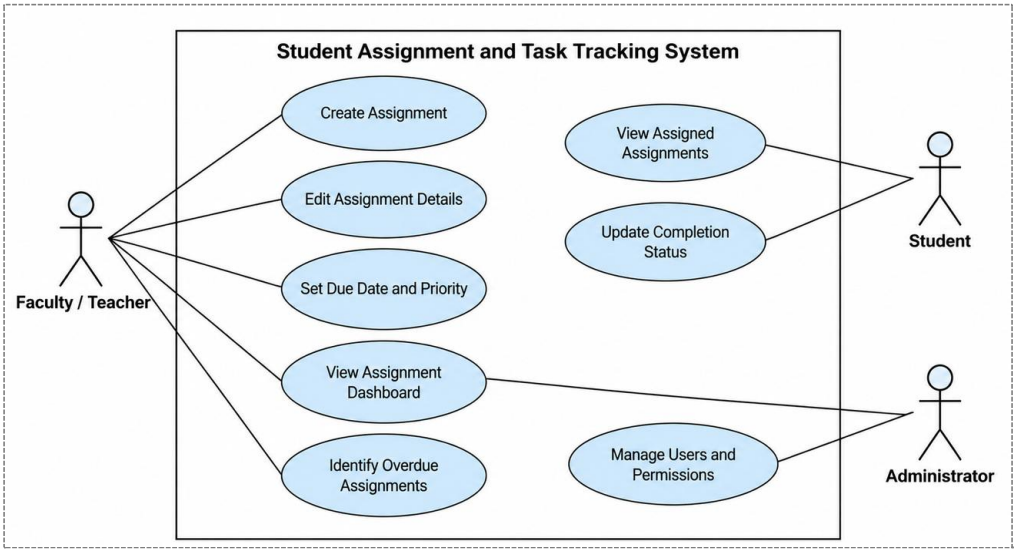



**Figure 3.1:** Use Case Diagram of the Student Assignment and Task Tracking System

Table 3.5: Use Case Description – Create and Track Assignments

| | |
|---|---|
| **Use Case ID** | UC-01 |
| **Use Case Name** | Create and Track Assignments |
| **Actor(s)** | Faculty / Teacher, Student |
| **Pre-condition** | The user must be logged in and have the appropriate access permissions. |
| **Main Flow** | 1. Faculty opens the Assignment Form. 2. Faculty enters the assignment title, subject, description, due date and priority. 3. The system validates the mandatory fields. 4. Faculty submits the assignment. 5. The system saves the assignment record. 6. The student views the assigned task and updates its completion status. |
| **Alternate Flow** | If mandatory fields are missing or invalid, the system displays a validation message and requests correction. |
| **Post-condition** | The assignment is stored and available for authorised users to view; its completion status can be tracked. |
| **Related FR** | FR-01, FR-02, FR-03, FR-04, FR-07 |

# CHAPTER 4
# SYSTEM DESIGN

## 4.1 System Architecture

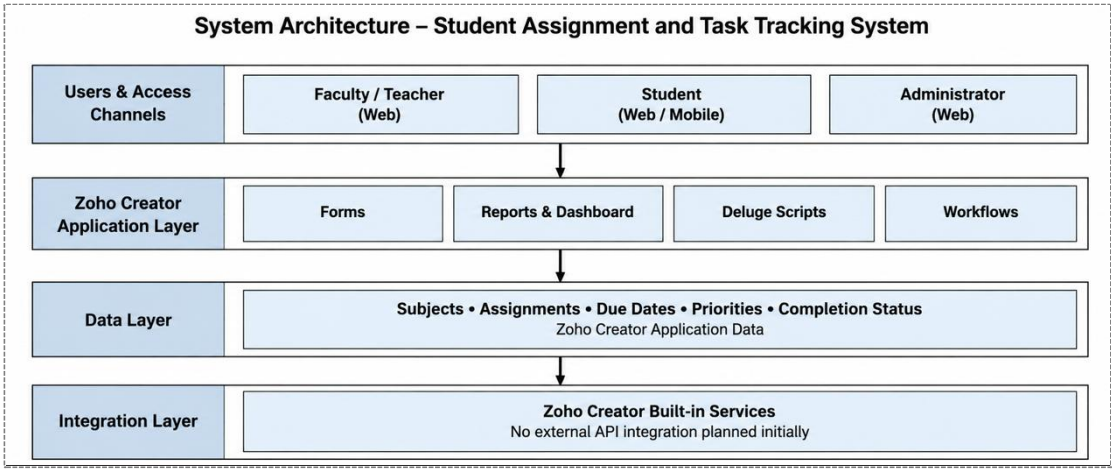



**Figure 4.1:** System Architecture of the Student Assignment and Task Tracking System

The system architecture consists of four layers. The User Access Layer represents the administrator using the Zoho Creator web application. The Zoho Creator Application Layer contains forms for entering subject, assignment and student task details, reports and a dashboard for monitoring progress, Deluge scripts for validation and business logic, and workflows for automating selected activities. The Data Layer stores subject details, assignment records, due dates, priorities and completion statuses within Zoho Creator. No external API integration is configured in the current implementation. These layers work together to provide a centralised system for managing and tracking academic assignments.

## 4.2 Data Model

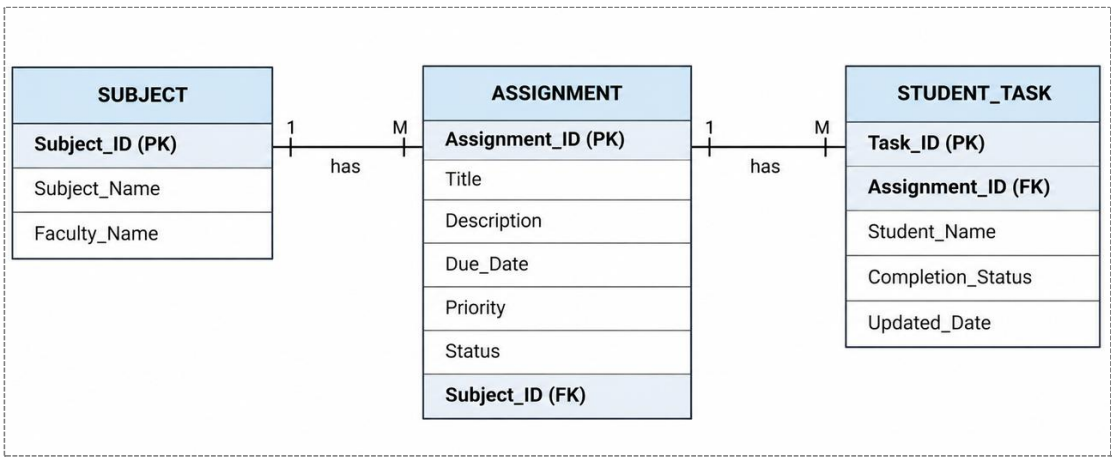



**Figure 4.2:** Entity–Relationship Diagram of the Student Assignment and Task Tracking System

**Table 4.1:** Entities and Relationships

| Entity (Form) | Key Fields | Related To | Relationship Type | Implemented Using |
|---|---|---|---|---|
| Subject | Subject ID, Subject Name, Faculty Name | Assignment | One-to-Many | Lookup field in Assignment |
| Assignment | Assignment ID, Title, Description, Due Date, Priority, Status, Subject ID | Subject, Student Task | One-to-Many with Student Task | Lookup fields |
| Student Task | Task ID, Assignment ID, Student Name, Completion Status, Updated Date | Assignment | Parent–Child / One-to-Many | Lookup field in Student Task |

## 4.3 Form Design

**Table 4.2:** Field Specification –Subject Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Subject ID | Subject_ID | Auto Number | Yes | Auto-generated | System generated |
| Subject Name | Subject_Name | Single Line | Yes | No duplicate values | Subject name |
| Faculty Name | Faculty_Name | Name | Yes | Required field | Faculty responsible |

Table 4.3: Field Specification – Assignment Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Assignment ID | Assignment_ID | Auto Number | Yes | Auto-generated | System generated |
| Title | Title | Single Line | Yes | Single Line | Assignment title |
| Description | Description | Multi Line | No | Optional | Assignment details |
| Subject | Subject | Lookup | Yes | Select an existing subject | Links to Subject Form |
| Due Date | Due Date | Date | Yes | Must be a valid date | Submission deadline |
| Priority | Priority | Drop Down | Yes | Default: Medium | High / Medium / Low |

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Status | Status | Drop Down | Yes | Default: Assigned | Assigned / In Progress / Completed |

**Table 4.4:** Field Specification –Student Task Form

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Task ID | Task ID | Auto Number | Yes | Auto-generated | System generated |
| Assignment | Assignment | Lookup | Yes | Select an existing assignment | Links to Assignment Form |
| Student Name | Student Name | Name | Yes | Required field | Student assigned to the task |
| Completion Status | Completion Status | Drop Down | Yes | Default: Not Started | Not Started / In Progress / Completed |
| Updated Date | Updated Date | Date-Time | No | Set during status update if configured | Tracks latest update |

### 4.4 Process and Workflow Design

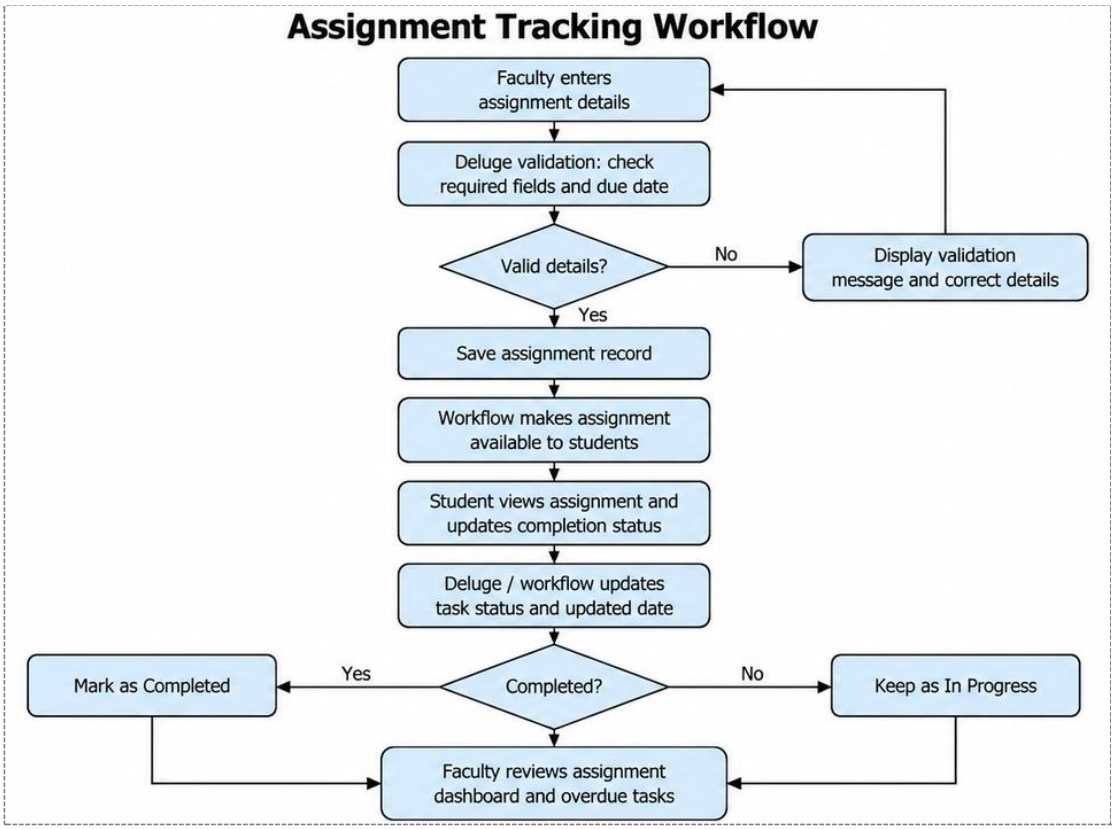


**Figure 4.3:** Workflow / Process Flow for Assignment Tracking

## 4.5 User Interface and Navigation Design

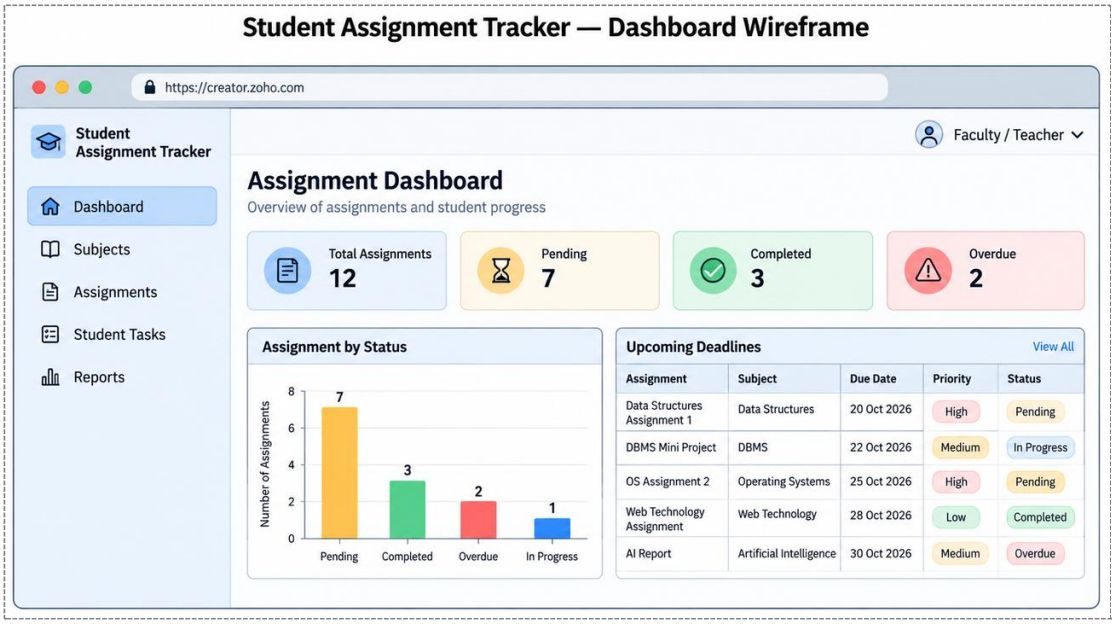



**Figure 4.4:** Navigation Structure and Wireframe of the Assignment Dashboard

## 4.6 Access Control Design

**Table 4.5:** Role–Permission Matrix

| Component | Administrator | Faculty / Teacher | Student |
|---|---|---|---|
| Form: Subject | A E V D | A E V | V |
| Form: Assignment | A E V D | A E V | V |
| Form: Student Task | A E V D | A E V | A E V (own) |
| Report: All Assignments | V | V | — |
| Report: Student Tasks | V | V | V (own) |
| Page: Assignment Dashboard | V | V | V (own tasks only) |

# CHAPTER 5
# IMPLEMENTATION

## 5.1 Application Setup

The Student Assignment and Task Tracking System was developed using Zoho Creator as a low-code application. The application manages subjects, assignments, due dates, priorities and student task progress. It contains Subject, Assignment and Student Task forms, reports and a dashboard. Deluge scripting and three form workflows implement due-date validation, automatic priority assignment and completion email notification. The application was built using the Creator application-building environment; the report does not claim that every component was programmed from scratch.

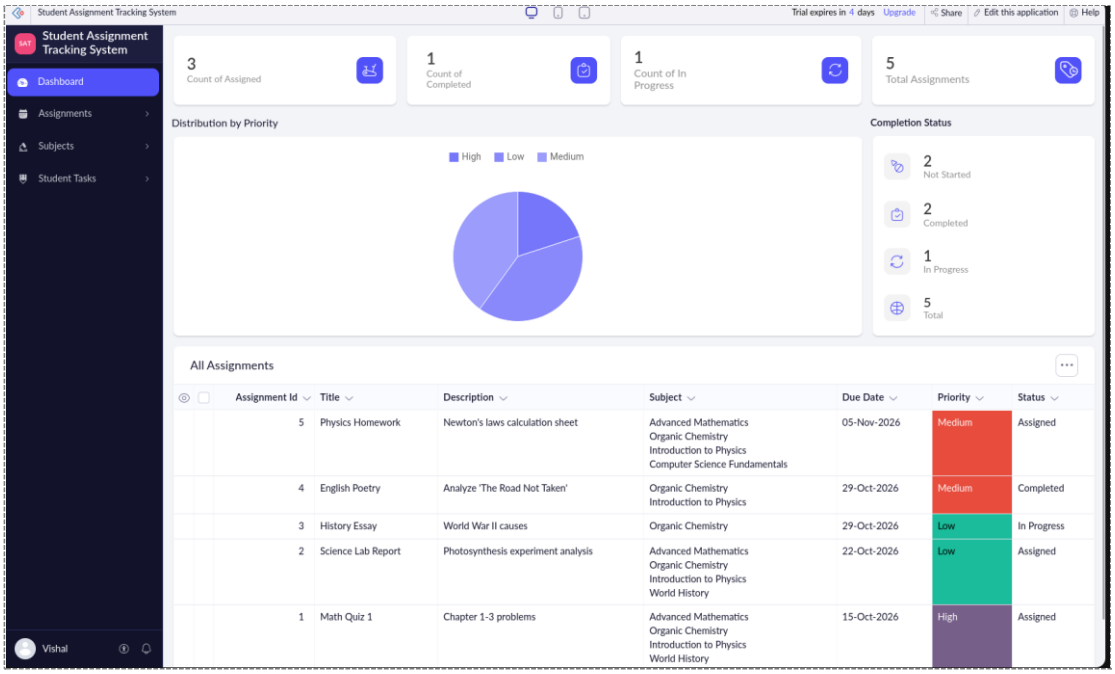



**Figure 5.1:** Application Home and Navigation Sections

## 5.2 Forms and Data Capture

The Student Assignment Tracking System uses three main forms: Subject, Assignment and Student Task. The Subject form captures subject details and faculty information. The Assignment form records assignment titles, descriptions, associated subjects, due dates, priorities and status. The Student Task form links an assignment to a student record and stores completion status and related task details. Lookup fields connect the Subject, Assignment and Student Task forms. The live forms and their actual field types should be treated as the source of truth for the field specifications in this chapter.

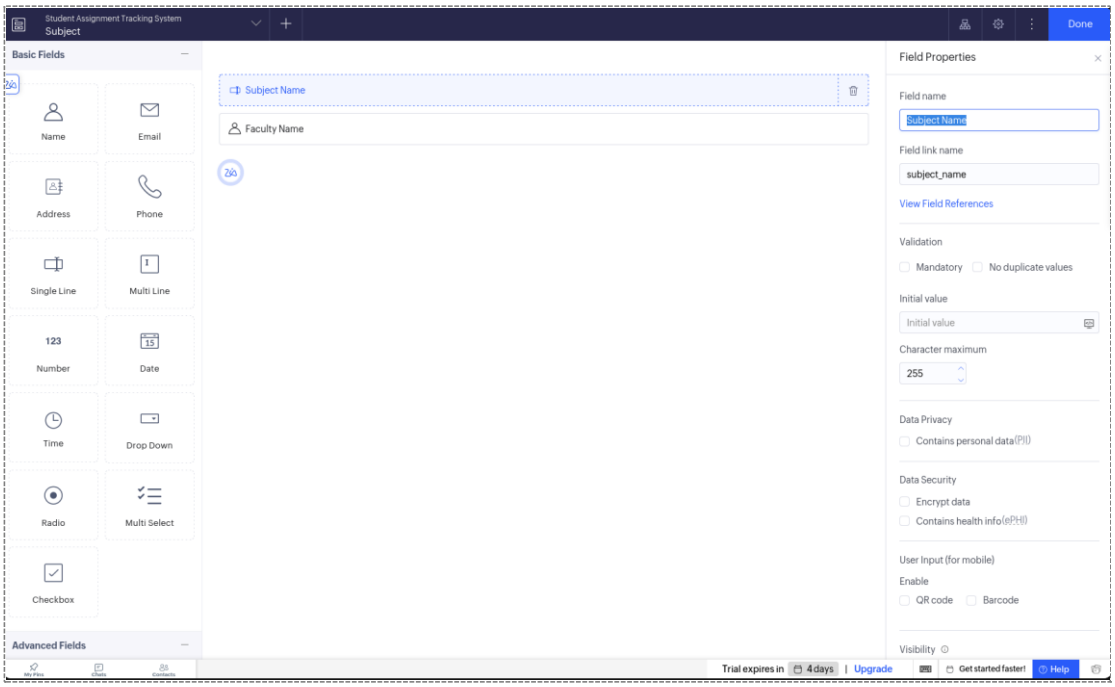



**Figure 5.2:** Subject Form in the Form Builder

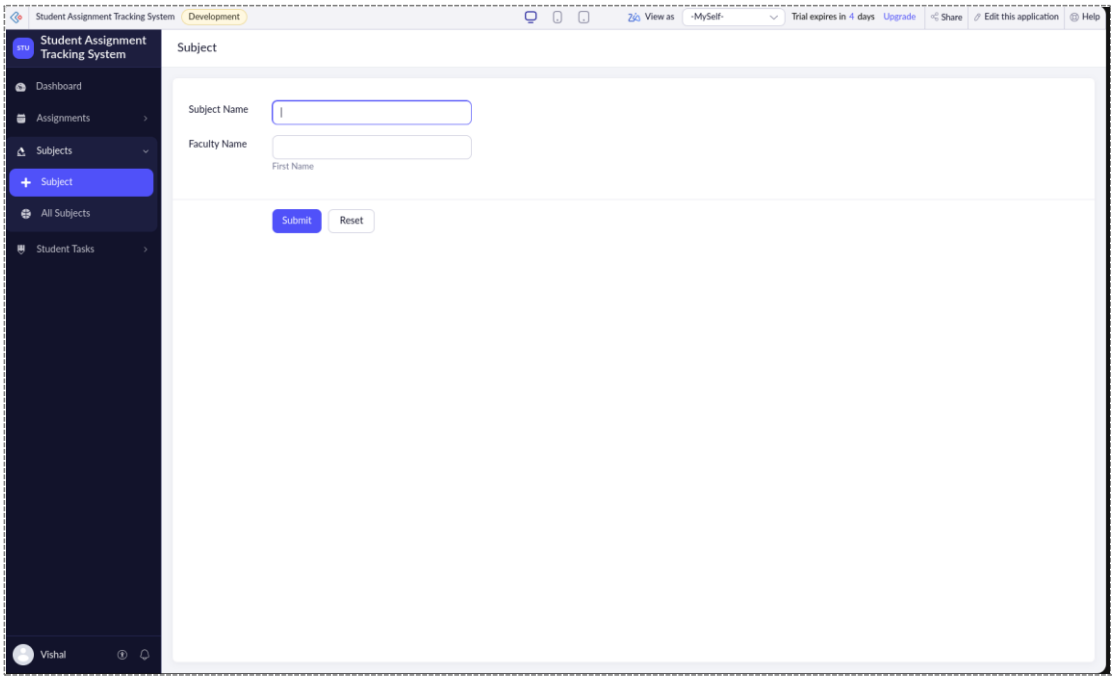



**Figure 5.3:** Subject Form in Live Mode

## 5.3 Reports

**Table 5.1:** Reports Implemented

| Report Name | Type | Source Form | Filter / Grouping / Sorting | Custom Actions | Roles |
|---|---|---|---|---|---|
| All Assignments | List | Assignment | Displays assignment title, subject, due date, priority and status | Default record actions | Administrator, Faculty |
| All Subjects | List | Subject | Displays subject name and faculty name | Default record actions | Administrator, Faculty |
| All Student Tasks | List | Student Task | Displays assignment, student name, completion status and updated date | Default record actions | Administrator, Faculty |
| Completion Statuses | List | Student Task | Organises task records by completion status | Default record actions | Administrator, Faculty, Student |
| Assignment Dashboard | Dashboard | Assignment | Displays assignment counts and priority distribution | Navigation to reports | Administrator, Faculty |

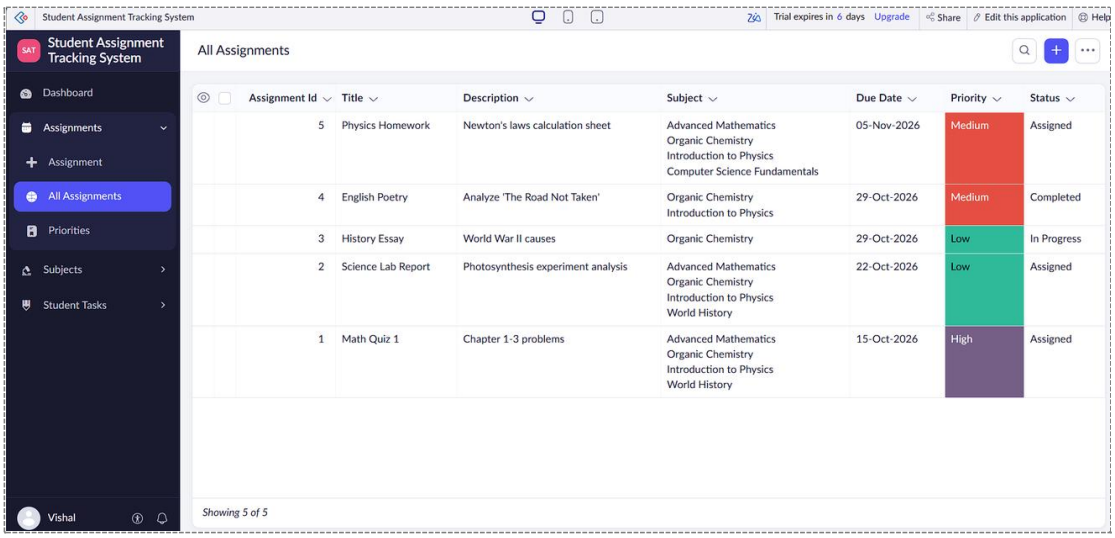



**Figure 5.4:** All Assignment Report

## 5.4 Pages and Dashboards

The Student Assignment Tracking System includes a dashboard that provides a consolidated view of assignment information and student task progress. It displays summary cards for assigned, completed and in-progress assignments, along with the total number of assignments. A priority distribution chart presents the distribution of assignments by High, Medium and Low priority. A completion status panel shows the counts of tasks in different stages, while the embedded All Assignments report displays

assignment titles, descriptions, subjects, due dates, priorities and statuses. The navigation menu provides access to the Dashboard, Assignments, Subjects and Student Tasks sections. These components provide the application administrator with a consolidated view of academic assignments and recorded task progress.

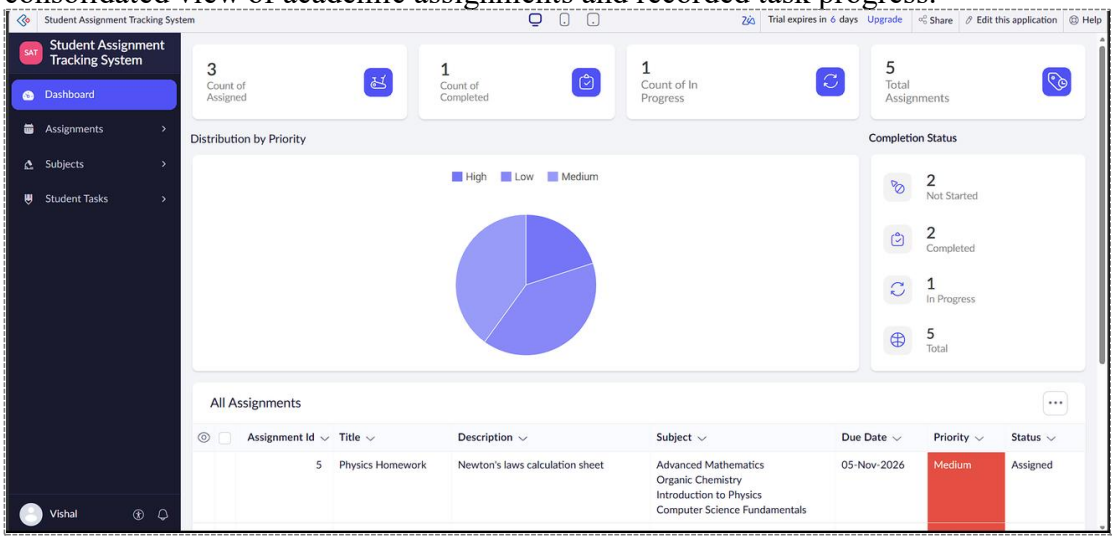



**Figure 5.5:** Assignment Tracking Dashboard

## 5.5 Deluge Scripting

**Table 5.2:** Deluge Scripts Implemented

| Script ID | Location / Trigger | Purpose | Deluge Concepts Used |
|---|---|---|---|
| DS-01 | Assignment → Created or Edited → User input of Due Date | Automatically sets priority to High when the due date is within the next three days. | Variables, date calculations, if conditions, field assignment |
| DS-02 | Assignment → Created or Edited → Validations on form submission | Checks whether the due date is entered, valid, and not in the past. | Null checks, date validation, conditional statements, alert, cancel submit |
| DS-03 | Student Task → Edited → User input of Completion Status | Sends an email notification to the student when the task status changes to Completed. | Record fetching, lookup-field access, conditional statements, string concatenation, email validation, sendmail |

***Code 5.1: Automatic Priority Assignment***

```
// DS-01 | Assignment | User input of Due Date
// Purpose: Set High priority for assignments due within 3 days.

inputDueDate = input.due_date;
todayDate = zoho.currentdate;

if(inputDueDate != null && !inputDueDate.isEmpty())
{
    daysBetween = todayDate.daysBetween(inputDueDate);

    // Set High priority for due dates 0-3 days away.
    if(daysBetween >= 0 && daysBetween <= 3)
    {
        input.priority = "High";
    }
}
```

The script reads the assignment due date and compares it with the current date. If the assignment is due today or within the next three days, its priority is automatically set to High. Due dates more than three days away do not trigger this priority change.

***Code 5.2: Assignment Due-Date Validation***

```
// DS-02 | Assignment | Validations on form submission
// Purpose: Validate the assignment due date.

if(input.due_date.isNull())
{
    alert "Due date is required. Please enter a valid due date.";
    cancel submit;
}

if(input.due_date.isDate())
{
    currentDate = zoho.currentdate;

    if(input.due_date < currentDate)
    {
        alert "Error: The due date cannot be in the past. Please
select a future date.";
        cancel submit;
    }
}
else
{
    alert "Error: Invalid due date format. Please enter a valid
date.";
    cancel submit;
}
```

This script checks the due-date field before the assignment is submitted. If the date is missing, invalid, or earlier than the current date, it displays an alert and cancels submission. Otherwise, the record can proceed through the normal submission process.

Code 5.3: Student Task Completion Email Notification

```
// DS-03 | Form: Student Task
// Trigger: Edited - User input of Completion Status
// Purpose: Notify the student when a task is completed.

studentTaskRecord = student_task[ID == input.ID];

if(studentTaskRecord.completion_status == "Completed")
{
    studentFirstName = studentTaskRecord.student_name.first_name;
    studentLastName = studentTaskRecord.student_name.last_name;
    assignmentTitle = studentTaskRecord.assignment.title;
    studentEmail = studentTaskRecord.Added_User.email;

    if(!studentEmail.isNull() && !studentEmail.isEmpty())
    {
        sendmail
        [
            from :zoho.loginuserid
            to :studentEmail
            subject :"Your Assignment is Completed: " +
assignmentTitle
            message :"Dear " + studentFirstName + " " +
studentLastName +
            ",\n\nYour assignment \"" + assignmentTitle +
            "\" has been marked as Completed.\n\nBest
regards,\nAssignment Tracking System"
        ]
    }
}
```

The workflow retrieves the student task record and checks whether its completion status is Completed. If so, it extracts the student's name and the associated assignment title, then obtains the student's email address from the Added_User field. When a valid email address is available, the script sends a notification email confirming that the assignment has been marked as completed.

## 5.6 Workflows and Automation

**Table 5.3:** Workflows Implemented

| Workflow Name | Type | Trigger / Event | Condition | Action(s) | Script Ref. |
|---|---|---|---|---|---|
| Auto-Assign Priority Based on Due Date | Form Workflow | Assignment created or edited; user input of Due Date | Due date is within 0–3 days from today | Set Priority to High | DS-01 |
| Validate Assignment Due Date | Form Workflow | Assignment created or edited; validation on form submission | Due date is missing, invalid, or in the past | Display alert and cancel submission | DS-02 |
| Notify Student on Task Completion | Form Workflow | Student Task edited; Completion Status input | Triggered when the completion status is edited | Executes one configured action to notify the student | DS-03 |

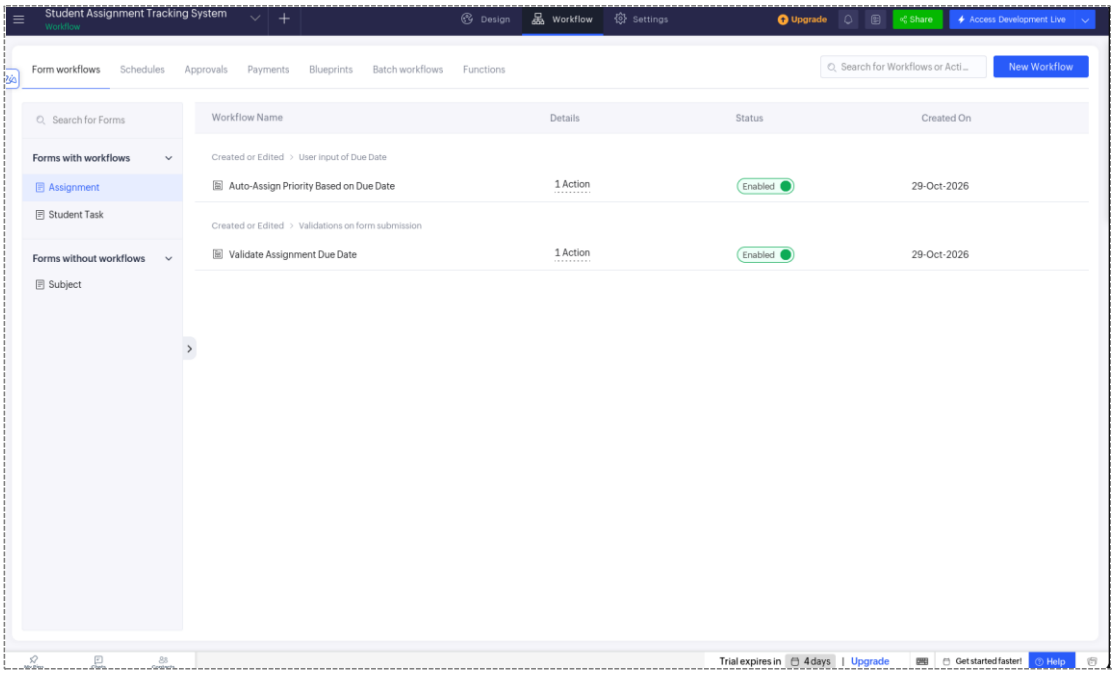



Figure 5.6: Assignment Workflow Configuration in Zoho Creator

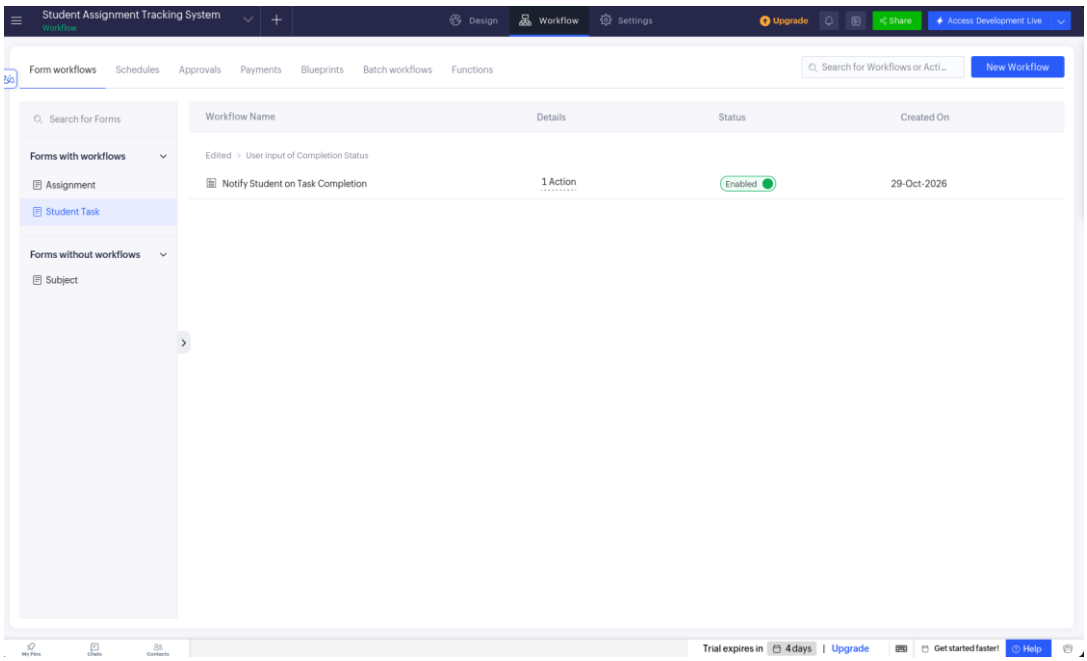



Figure 5.6: Assignment Workflow Configuration in Zoho Creator

## 5.7 Integrations

**Table 5.4:** Integrations Implemented

| Integration | Service / Endpoint | Method | Authentication | Data Exchanged / Purpose |
|---|---|---|---|---|
| Email Notification | Zoho Creator sendmail task | Built-in Deluge email task | Zoho Creator account | Sends a completion notification containing the assignment title and student name. |

Built-in Email Notification (documented under Code 5.3)

> No external integration script was required. The completion email is a built-in Zoho Creator sendmail action configured in the Student Task workflow and documented under Code 5.3.

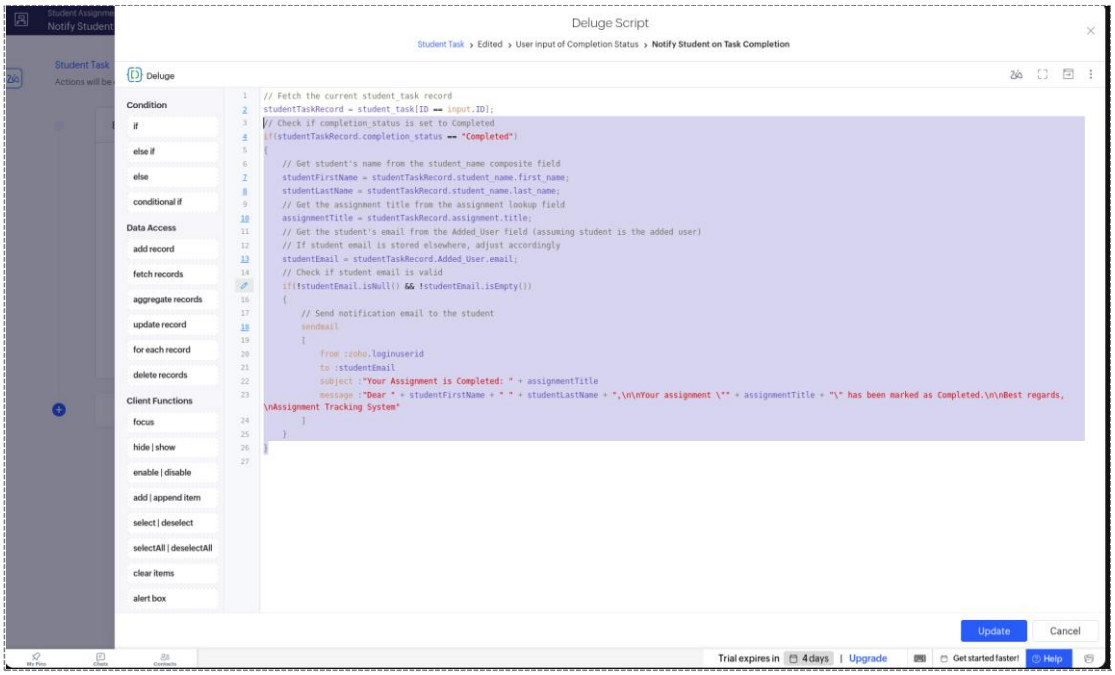



Figure 5.7: Student Task Completion Notification Workflow

## 5.8 User Management and Role-Based Access

Zoho Creator provides application-level user and access-management capabilities. The current application is demonstrated using its configured administrator access. Separate Faculty and Student permission profiles and record-level restrictions were not confirmed as implemented. Therefore, the roles described in the requirements and design chapters should be treated as intended user roles unless the corresponding permissions are configured and demonstrated.

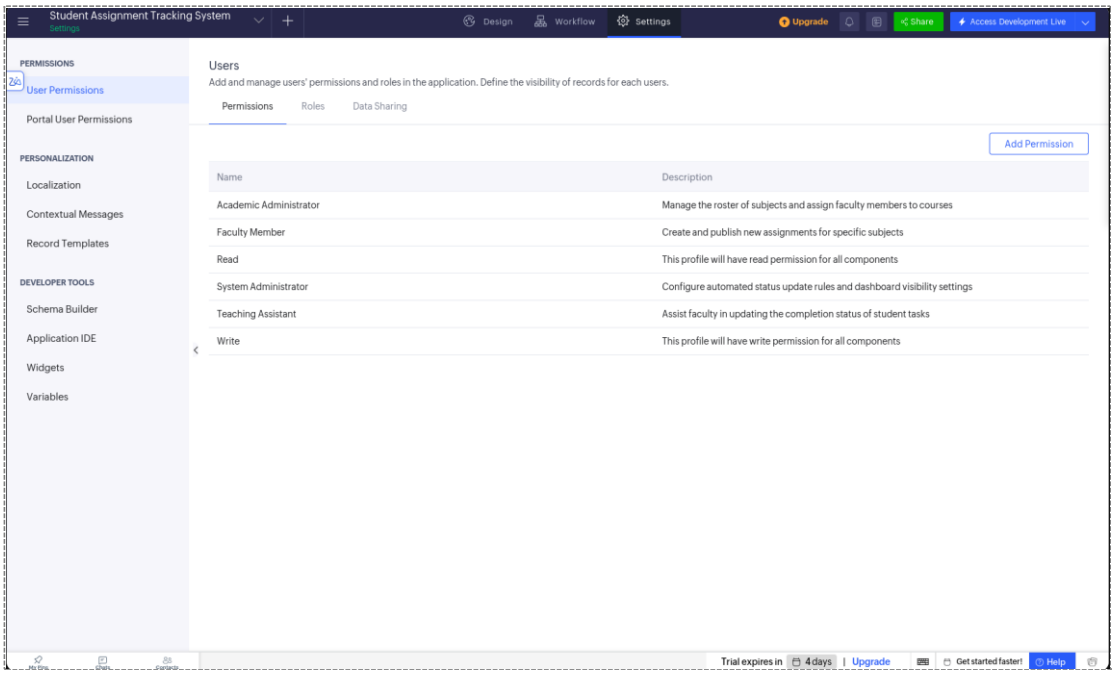



**Figure 5.8:** Roles and Permission Profiles Configuration

## 5.9 AI and Analytics Features

The Student Assignment Tracking System provides analytical features through its dashboard in Zoho Creator. Summary cards display the total number of assignments and counts of assignments by status, including Assigned, In Progress and Completed. A priority distribution chart presents the distribution of assignments across High, Medium and Low priority levels. These visualisations help users understand the current assignment workload and monitor completion progress. The application also uses a Deluge workflow to automatically assign High priority to assignments due within the next three days. This is rule-based automation, not a machine-learning model. No separate AI service or Zia feature was integrated.

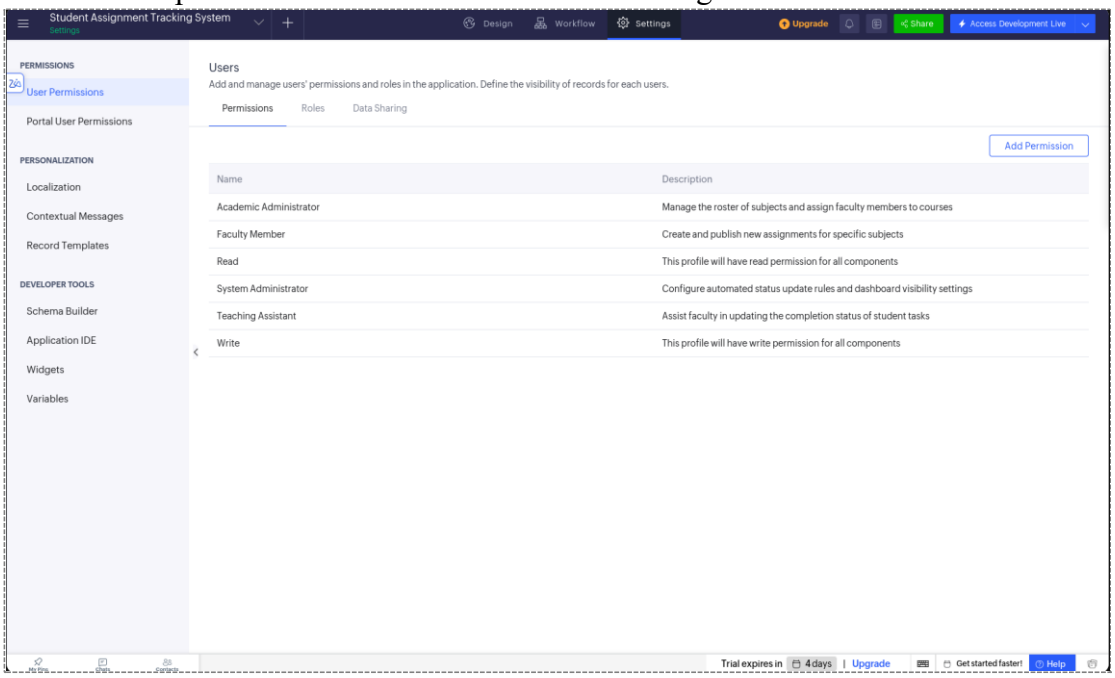
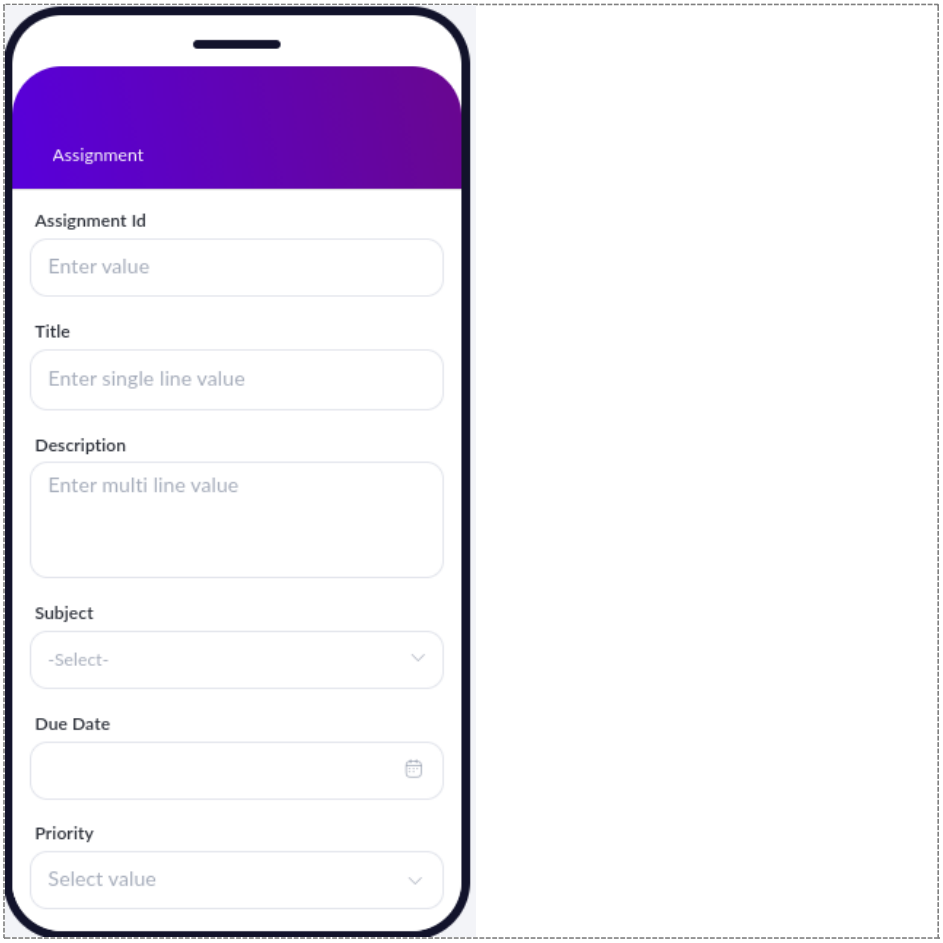


**Figure 5.9:** Analytics Features in Use

## 5.10 Pro-Code Extension

The application was developed using Zoho Creator's low-code environment and Deluge scripting to implement assignment priority automation, due-date validation and student completion email notifications. Additional pro-code extensions, such as custom JavaScript widgets, external REST API integrations and Zoho Catalyst functions, were not used in the current implementation. These extensions could be considered in future versions to support advanced analytics, integration with external academic systems and customised user interface components. The current low-code approach was sufficient to implement the core assignment tracking features within the project scope.

## 5.11 Mobile and Portal Access




Mobile and portal access were not implemented or demonstrated in the current version of the application.

# CHAPTER 6
# TESTING AND DEBUGGING

## 6.1 Testing Strategy

The Student Assignment Tracking System was reviewed using sample subject, assignment and student task records. Form-level checks covered record creation and report visibility. The three configured workflows were reviewed for their triggers, conditions and Deluge actions. The priority assignment, due-date validation and completion email behaviour require direct execution tests before they can be marked as passed. Dashboard and report visibility was confirmed in the application. Test results below distinguish demonstrated behaviour from workflows that are configured but not yet verified.

## 6.2 Test Cases

**Table 6.1:** Test Cases

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-01 | FR-01 | Submit Subject form with mandatory fields | Enter subject name and faculty name; submit. | Subject record created successfully. | Subject record displayed in All Subjects report. | Pass |
| TC-02 | FR-01 | Submit Assignment form without due date | Leave due date blank and submit. | Validation message displayed; submission cancelled. | Validation workflow is configured; execution with an empty due date has not been recorded. | Not Tested |
| TC-03 | FR-02 | Set assignment due within three days | Enter a due date within three days and save. | Priority automatically set to High. | Priority workflow is enabled and configured; output has not been recorded in a test. | Not Tested |
| TC-04 | FR-02 | Submit assignment with a past due date | Select a past date and submit. | Validation alert displayed; submission cancelled. | Past-date validation is configured; execution with a past date has not been recorded. | Not Tested |

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC-05 | FR-03 | Mark Student Task as Completed | Edit a task and set status to Completed. | Completion notification workflow sends an email if a valid recipient email is available. | Notification workflow is enabled; email delivery has not been verified. | Not Tested |
| TC-06 | FR-05 | View assignment dashboard and reports | Open the dashboard and All Assignments report. | Assignment statistics and records are displayed. | Dashboard and report are visible in the application. | Pass |

### 6.3 Debugging Deluge Scripts

Debugging was performed by reviewing the Deluge scripts configured in Zoho Creator and checking the workflow triggers, field link names, conditions and actions. The due-date validation script was reviewed to ensure that missing or past dates display an alert and cancel form submission. The priority workflow was checked for its date comparison logic, while the Student Task workflow was reviewed to verify the completion-status condition and email notification logic. Error messages during form submission and workflow execution can be used to identify incorrect field references or invalid conditions. The scripts should be tested with valid and invalid inputs to confirm their behaviour.

### 6.4 Defect Log

**Table 6.2:** Defect Log

| Bug ID | Description | Root Cause | Fix Applied | Status |
|---|---|---|---|---|
| BUG-01 | Due-date validation may not handle an empty or invalid date correctly. | Date-field checks in the validation script require testing against the actual Zoho Creator field type. | Review and test null checks, date validation and submission cancellation. | Under Testing |
| BUG-02 | Completion notification may be sent to the record creator instead of the intended student. | The script retrieves the email from Added_User.email. | Verify the recipient field and use the student's actual email field if different. | Under Testing |

### 6.5 User Acceptance Testing

**Table 6.3:** User Acceptance Feedback

| User | Role Tested | Feedback | Action Taken |
|---|---|---|---|
| User 1 | Faculty / Administrator | The dashboard provides a clear overview of assignments, priorities and completion status. | Reviewed the dashboard and assignment reports for clarity. |

| User | Role Tested | Feedback | Action Taken |
|---|---|---|---|
| User 2 | Student | The assignment list makes it easier to view due dates, priorities and task completion status. | Reviewed the task list and completion-status display. |

# CHAPTER 7
# DEPLOYMENT, SECURITY AND GOVERNANCE

## 7.1 Environments and Deployment Strategy

The Student Assignment Tracking System was developed and configured in Zoho Creator. A Production version is shown in the application environment view, and the Dashboard page has a published URL recorded in the Project at a Glance section. A separate Stage environment was not configured. Before public demonstration, the published URL should be tested in a private browser window and access to sample records should be reviewed. The deployment checklist below records only items supported by the current implementation.

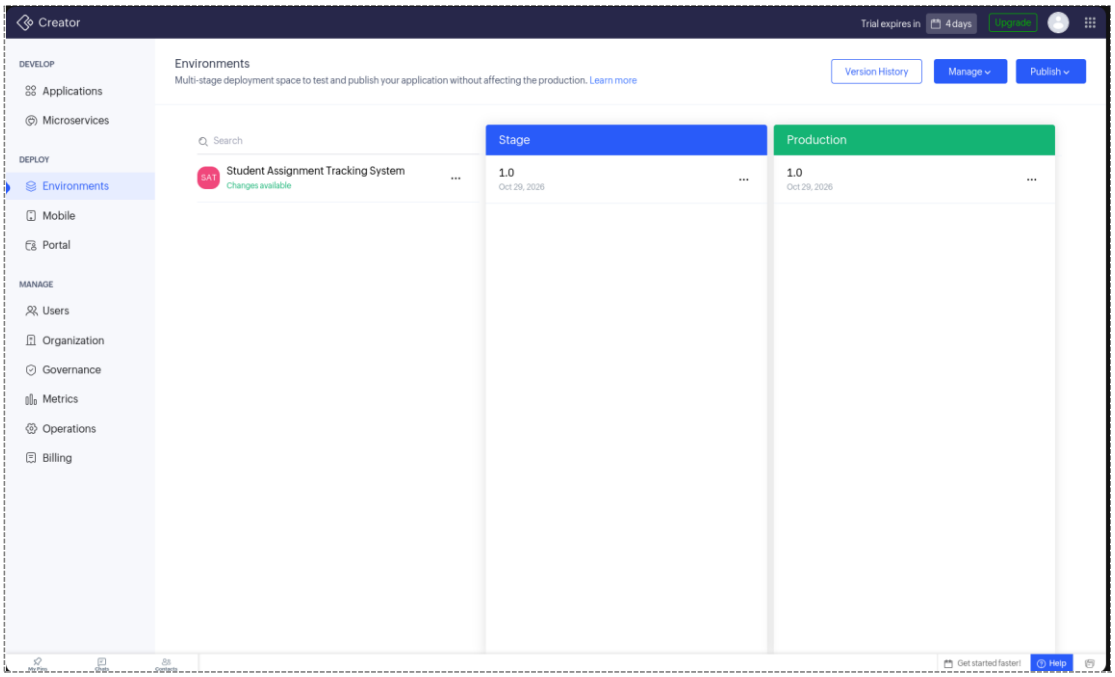



**Figure 7.1:** Environments

**Table 7.1:** Deployment Checklist

| S.No. | Checklist Item | Done (✓) |
|---|---|---|
| 1 | All test cases executed in Development / Stage | Not all tests verified |
| 2 | Sample/test data removed or isolated | Verify before public use |
| 3 | Roles, permissions and users configured in Production | Not configured / not verified |
| 4 | Connections authorised in Production | N/A |
| 5 | Schedules and workflows enabled | ✓ (3 form workflows) |

| S.No. | Checklist Item | Done (✓) |
|---|---|---|
| 6 | Application published and URL verified | ✓ (Dashboard URL recorded; verify public access) |
| 7 | Version / release notes recorded | Not recorded |

## 7.2 Performance Optimisation

Table 7.2: Performance Optimisation Activities

| Optimisation Applied | Component | Evidence / Measurement | Result | Status |
|---|---|---|---|---|
| Used criteria-based record lookup to retrieve the required Student Task record | DS-03 – Completion Notification | No baseline measured | Qualitative configuration only | Not quantitatively tested |
| Displayed assignment records through the relevant report | All Assignments report | No baseline measured | Qualitative configuration only | Not quantitatively tested |
| Used dashboard summary cards and a priority chart to present assignment information | Student Assignment Dashboard | No baseline measured | Qualitative configuration only | Not quantitatively tested |

## 7.3 Security Features

**Table 7.3:** Security Controls Implemented

| Security / Control Area | Where Applied | Status / Purpose |
|---|---|---|
| Zoho Creator account authentication | Application access | Platform-level sign-in; access depends on account configuration |
| Administrator access | Application management | Administrator access is used for application configuration |
| Due-date validation | Assignment form workflow | Data-quality control; rejects missing or past due dates when executed |
| Conditional completion email | Student Task workflow | Sends notification only when completion status is Completed and an email address is available |
| Separate Faculty / Student profiles | Application permissions | Not configured in the current implementation |

### 7.4 Governance and Administration

The Student Assignment Tracking System is managed through Zoho Creator's application management interface. The application administrator can manage application users, review application configurations and maintain the forms, reports and workflows used for assignment tracking. The three configured workflows support due-date validation, automatic priority assignment and email notification when a student task is marked as completed.

Sample subject, assignment and student task records are used to demonstrate the application's functionality. Before production use, user permissions should be reviewed to ensure that users can access only the information and functions required for their roles. Workflow behaviour should also be tested, and changes to the application should be reviewed before release. These practices help maintain data accuracy, controlled access and consistent application behaviour.

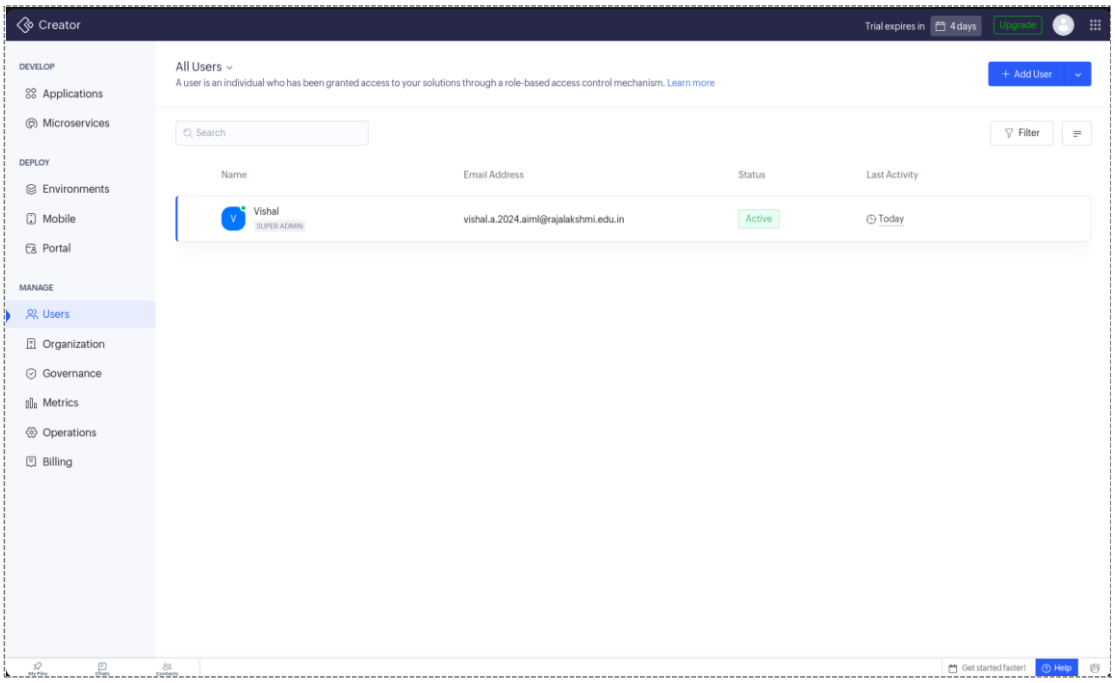



**Figure 7.2:** Administration Settings

# CHAPTER 8

# RESULTS AND DISCUSSION

## 8.1 Application Output

The Student Assignment Tracking System was demonstrated using sample subject, assignment and student task records. The scenario illustrates how assignment information is entered, monitored and updated through the application's forms, reports and dashboard.

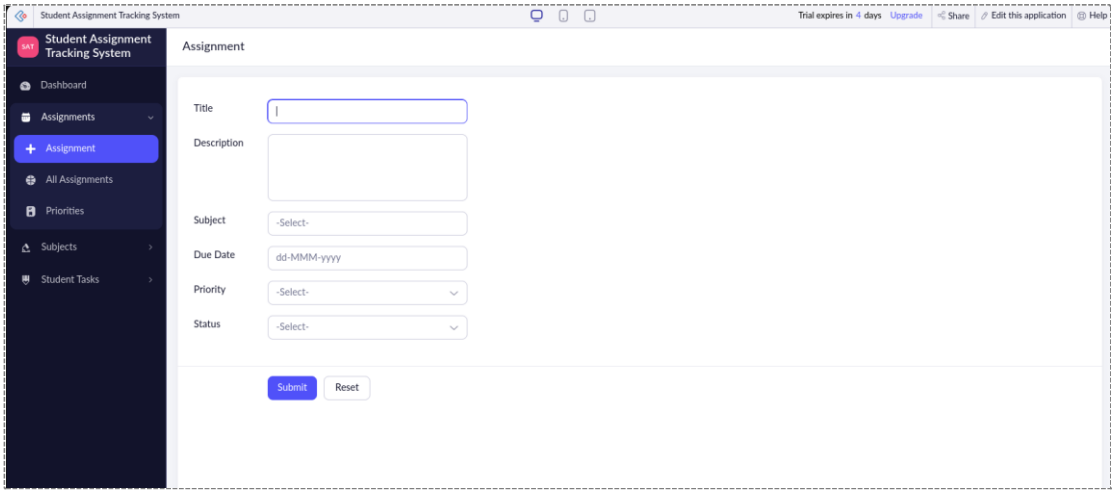



**Figure 8.1:** Scenario Step 1 – Creating an Assignment

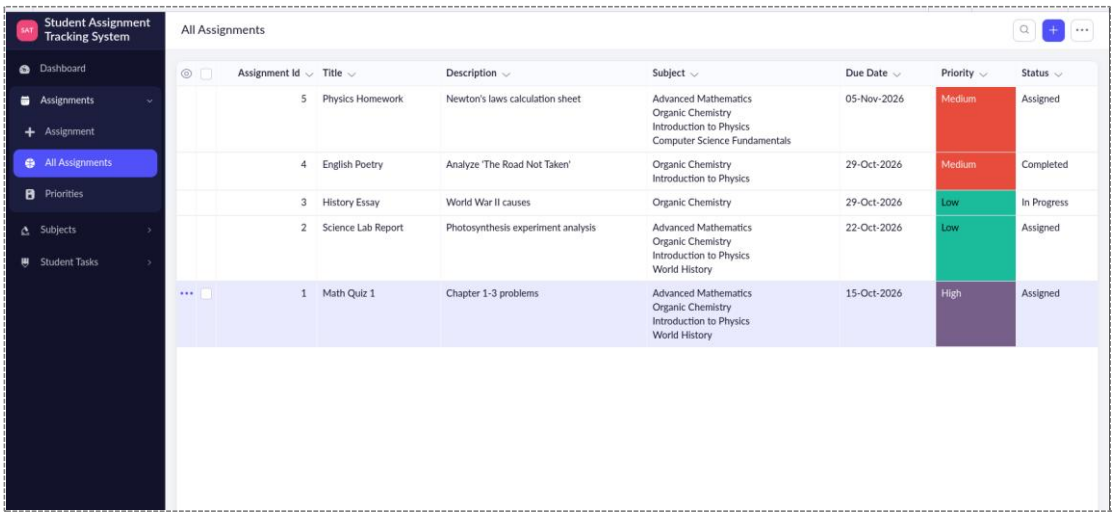



**Figure 8.2:** Scenario Step 2 – Viewing and Updating Assignment Status

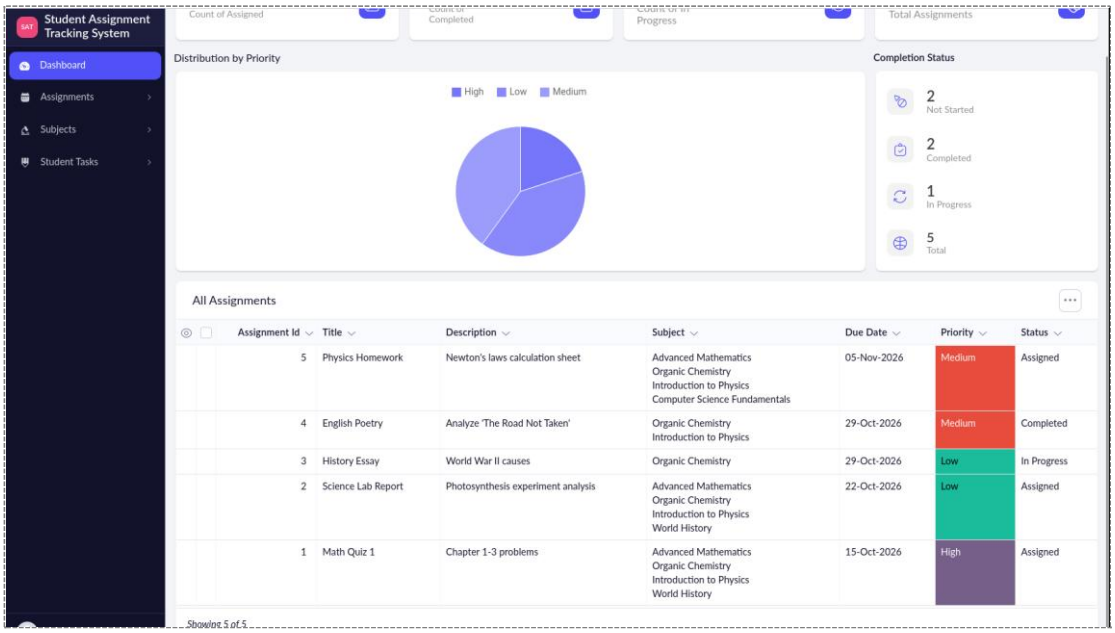



**Figure 8.3:** Scenario Step 3 – Reviewing Assignment Progress on the Dashboard

## 8.2 Achievement of Objectives

**Table 8.1:** Mapping of Objectives to Outcomes

| Objective (Sec. 1.3) | Implemented Through | Outcome / Evidence |
|---|---|---|
| Objective 1: Design forms and a data model for subjects, assignments, deadlines, priorities and completion status. | Sections 5.2 and 5.3 – Forms and Reports | Subject, Assignment and Student Task forms, with corresponding reports for managing records. |
| Objective 2: Automate deadline validation, priority assignment and task completion notifications. | Sections 5.5 and 5.6 – Deluge Scripting and Workflows | Three workflows configured for due-date validation, automatic priority assignment and completion email notifications. |
| Objective 3: Provide reports and a dashboard to monitor assignment progress. | Sections 5.3 and 5.4 – Reports and Dashboard | All Assignments report and dashboard displaying assignment summary cards, priority distribution and completion-status information. |

## 8.3 Comparison with the Existing System

**Table 8.2:** Existing System versus Proposed Application

| Parameter | Existing System | Proposed Application | Improvement |
|---|---|---|---|
| Time to process one assignment | Details recorded and tracked manually across different sources | Assignment details entered and viewed through Zoho Creator forms and reports | More organised assignment management |

| Parameter | Existing System | Proposed Application | Improvement |
|---|---|---|---|
| Manual steps | Students manually track deadlines and update task status | Due-date validation and automatic priority assignment are configured through workflows | Reduces some repetitive manual work |
| Report generation | Assignment information gathered from notebooks or separate records | All Assignments, All Subjects and All Student Tasks reports are available | Centralised viewing of assignment information |
| Data accessibility | Information may be scattered across different sources | Subject, assignment and task records are maintained in one application | Easier access to consolidated information |

### 8.4 Limitations of the Low-Code Approach

Zoho Creator simplifies application development through visual forms, reports, dashboards and workflow automation, but it has certain limitations in UI customisation and the implementation of complex business logic. Advanced features may require additional Deluge scripting, while external API integrations may involve extra configuration and authentication. The application also depends on the Zoho Creator platform, which can limit portability to other platforms. These limitations were addressed by using built-in components and Deluge workflows to implement the core features, including due-date validation, automatic priority assignment, completion notifications and dashboard reporting. Performance with large datasets and multiple users was not evaluated in the current project.

### 8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The Student Assignment Tracking System demonstrates the use of no-code, low-code and potential pro-code development approaches. No-code features were used to create forms, reports, dashboard components and charts through Zoho Creator's visual interface. Low-code development was applied through Deluge scripting to validate assignment due dates, assign priorities automatically and send email notifications when tasks are completed. Pro-code technologies, such as custom JavaScript components, external API integrations and a separately developed backend, could provide greater flexibility and advanced functionality, but were not used in this project. Overall, Zoho Creator provided a suitable balance between development speed and functionality for the project's requirements, while a pro-code approach could be considered for future

enhancements involving complex logic, advanced analytics or extensive external integrations.

# CHAPTER 9
# CONCLUSION AND FUTURE ENHANCEMENTS

## 9.1 Conclusion

The Student Assignment Tracking System was developed using Zoho Creator to simplify the management of academic subjects, assignments and student tasks. The application includes Subject, Assignment and Student Task forms, reports for viewing records, and a dashboard displaying assignment statistics, priority distribution and completion status. Three workflows were configured to validate assignment due dates, automatically assign High priority to assignments with approaching deadlines, and send email notifications when student tasks are marked as completed. Workflow configuration is documented; individual workflow execution and email delivery should only be described as tested when verified.

The project demonstrates how a low-code platform can be used to develop a practical academic task-management application with reduced coding effort. The integration of visual application components and Deluge scripting provides an organised way to record assignment details, monitor deadlines and review progress. The project also provided practical experience in application design, workflow automation, reporting and dashboard development using Zoho Creator.

## 9.2 Future Enhancements

- AI-Based Predictive Analytics: Integrate Zoho Zia or machine learning models to predict assignment delays, analyse completion patterns and recommend task priorities.
- Calendar and LMS Integration: Integrate Google Calendar and college Learning Management Systems to automatically synchronise assignment deadlines and send reminders.
- Advanced Analytics and Microservices: Use Zoho Catalyst and custom microservices to support advanced analytics, process larger datasets and generate detailed student performance reports.

## 9.3 Learning Outcomes and Career Relevance

This project provided practical experience in low-code application development using Zoho Creator, including form design, report creation, dashboard configuration and Deluge scripting. It improved my understanding of data management, workflow

automation, validation logic, notification handling and application administration. The project also demonstrated how low-code platforms can reduce development time while supporting useful business and academic applications. These skills are relevant to career paths such as low-code application developer, citizen developer, automation engineer and solutions consultant, and provide a foundation for exploring pro-code development and advanced application integrations in future projects.

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# APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

**Table A.1:** Feature Coverage Checklist

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| I | Application created & sections/menu organised | Y | Student Assignment and Task Tracking System | 5.1 | Fig. 5.1 |
| II | Forms with ≥ 8 distinct field types | Y | Subject, Assignment, Student Task | 5.2 | Figs. 5.2–5.3 |
| II | Lookup field (relationship between forms) | Y | Assignment → Subject; Student Task → Assignment | 4.2, 5.2 | Fig. 4.2 |
| II | Subform (parent–child data) | N | Not implemented | 4.2, 5.2 | - |
| II | Field validations / layout rules | Y | Assignment form – due-date validation | 5.2 | Code 5.1 |
| II | List report with filter, grouping, custom action | N | List reports implemented; advanced features not confirmed | 5.3 | Fig. 5.4 |
| II | Kanban / Calendar / Timeline / Map report | N | Not implemented | 5.3 | - |
| II | Pivot table / chart report | Y | Dashboard – priority distribution chart | 5.3 | Fig 5.5 |
| II | Page / dashboard built with UI Builder | Y | Student Assignment Dashboard | 5.4 | Fig. 5.5 |
| II | Deluge – form events (validate / success / field actions) | Y | Assignment and Student Task workflows | 5.5 | Codes 5.1 - 5.3 |
| II | Deluge – functions, fetch, loops, collections | Y | Student Task workflow – criteria-based record fetch | 5.5 | Code 5.3 |
| II | Email / in-app notifications | Y | Notify Student on Task Completion | 5.5, 5.6 | Code 5.3 |
| II | Form workflow | Y | Three configured form workflows | 5.6 | Fig. 5.6 |
| II | Scheduled workflow | N | Not implemented | 5.6 | - |

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| II | Approval process | N | Not implemented | 5.6 | - |
| II | Blueprint | N | Not implemented | 5.6 | - |
| II | Batch / report workflow | N | Not implemented | 5.6 | - |
| II | Testing & debugging evidence | N | Forms, reports and workflow testing | 6 | - |
| III | Integration with a Zoho service | N | No separate integration configured | 5.7 | - |
| III | Connection / REST API call (invokeurl) | N | Not implemented | 5.7 | - |
| III | Webhook or Creator REST API | N | Not implemented | 5.7 | - |
| III | Users, roles and permission profiles | N | Only Administrator access confirmed | 5.8 | - |
| III | Record-level access criteria / customer portal | N | Not implemented | 5.8 | - |
| III | Zia / AI feature | N | No separate AI feature integrated | 5.9 | - |
| III | Analytics (charts / Zoho Analytics) | Y | Dashboard – priority chart and summary cards | 5.9 | Fig. 5.5 |
| IV | Performance optimisation applied & measured | N | No quantitative performance measurement | 7.2 | Table 7.2 |
| IV | Deployment across environments | N | Production version 1.0; Stage not configured | 7.1 | Fig. 7.1 |
| IV | Security controls (encryption, OAuth, permissions) | N | Advanced custom security controls not configured | 7.3 | Table 7.3 |
| IV | Governance (audit trail, administration) | N | No audit-trail implementation confirmed | 7.4 | - |
| V | Pro-code extension (Catalyst / widgets / custom functions) | N | Not implemented | 5.10 | - |
| V | Low-code limitations & spectrum analysis | N | Discussion of limitations and development spectrum | 8.4, 8.5 | - |
| — | Mobile app / portal access | N | Not implemented or demonstrated | 5.11 | - |

Total features marked implemented: 11 / 32.

# APPENDIX B – COURSE OUTCOME MAPPING

**Table B.1:** Mapping of Course Outcomes to Report Chapters

| CO | Course Outcome Statement | Evidenced In | Self-Assessed Level | Examiner Level |
|---|---|---|---|---|
| CO1 | Explain the principles, architecture, and business value of low-code application development using Zoho Creator. | Ch. 1, 2 | Review By student | |
| CO2 | Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting. | Ch. 4, 5 (5.1–5.6), 6 | Review By student | |
| CO3 | Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security. | Ch. 4.6, 5.7–5.9 | Review By student | |
| CO4 | Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices. | Ch. 7 | Review By student | |
| CO5 | Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer. | Ch. 5.9, 5.10, 8.4, 8.5, 9.3 | Review By student | |

# APPENDIX C – COMPLETE DELUGE SOURCE CODE

***C.1*** Auto-Assign Priority Based on Due Date

```
// Script ID : DS-01
// Location  : Assignment - User Input of Due Date
// Purpose   : Automatically assign priority based on due date

// Get the due date from user input
inputDueDate = input.due_date;

// Get today's date
todayDate = zoho.currentdate;

// Calculate the number of days between today and the due date
if(inputDueDate != null && !inputDueDate.isEmpty())
{
    daysBetween = todayDate.daysBetween(inputDueDate);

    // If due date is within 3 days, set priority to High
    if(daysBetween >= 0 && daysBetween <= 3)
    {
        input.priority = "High";
    }
}
```

### *C.2* Validate Assignment Due Date

```
// Script ID : DS-02
// Location  : Assignment - Validation on Form Submission
// Purpose   : Validate assignment due date

// Check if due_date field exists and is not null
if(input.due_date.isNull())
{
    alert "Due date is required. Please enter a valid date.";
    cancel submit;
}

// Check if due date is in the past
if(input.due_date.isDate())
{
    currentDate = zoho.currentdate;

    if(input.due_date < currentDate)
    {
        alert "Error: The due date cannot be in the past.
Please select a future date.";
        cancel submit;
    }
}
else
{
    alert "Error: Invalid due date format. Please enter a valid
date.";
    cancel submit;
}
```

### *C.3* Notify Student on Task Completion

```
// Script ID : DS-03
// Location  : Student Task - User Input of Completion Status
// Purpose   : Send email notification when task is completed

// Fetch the current student_task record
studentTaskRecord = student_task[ID == input.ID];

// Check if completion_status is set to Completed
if(studentTaskRecord.completion_status == "Completed")
{
    // Get student's name from the student_name composite field
    studentFirstName =
studentTaskRecord.student_name.first_name;
    studentLastName = studentTaskRecord.student_name.last_name;

    // Get the assignment title from the assignment lookup field
    assignmentTitle = studentTaskRecord.assignment.title;

    // Get the student's email from the Added_User field
    studentEmail = studentTaskRecord.Added_User.email;

    // Check if student email is valid
    if(!studentEmail.isNull() && !studentEmail.isEmpty())
```

```
    {
        // Send notification email to the student
        sendmail
        [
            from :zoho.loginuserid
            to :studentEmail
            subject :"Your Assignment is Completed: " +
assignmentTitle
            message :"Dear " + studentFirstName + " " +
studentLastName + ",\n\nYour assignment \"" + assignmentTitle +
"\" has been marked as Completed.\n\nBest regards,\nAssignment
Tracking System"
        ]
    }
}
```

# APPENDIX D – USER MANUAL

## D.1 Accessing the Application

- Open the published Student Assignment Tracking System URL in a web browser.
- Sign in using the authorised Zoho account if authentication is required.
- After accessing the application, open the dashboard from the application navigation menu.
- Use the available forms and reports to manage and view assignment information.

## D.2 Instructions for Administrator

- Step 1 – Manage Subjects: Open the Subject form, enter the subject name and faculty details, and submit the form to create a subject record.
- Step 2 – Create Assignments: Open the Assignment form, enter the assignment title, description, subject, due date, priority and status, then submit the form.
- Step 3 – Review Assignments: Open the All Assignments report to view and manage the available assignment records.
- Step 4 – Monitor Progress: Open the dashboard to review assignment counts, priority distribution and completion status.

## D.3 Instructions for Student

- Step 1 – View Assignments: Open the All Assignments report to view assignment titles, subjects, due dates, priorities and statuses.
- Step 2 – Update Task Status: Open the Student Task form, select the relevant assignment, enter the student details and update the completion status.
- Step 3 – Review Progress: Open the dashboard or available reports to review assignment progress.

# APPENDIX E – EVALUATION RUBRIC

*(For Examiner Use Only)*

| Student Name | A VISHAL | Register No. | 241501247 |
|---|---|---|---|
| Project Title | Student Assignment and Tracking System | Date | |

**Table E.1:** Evaluation Rubric for the Individual Project

| S.No. | Criterion | Indicators | Max. Marks | Marks Awarded |
|---|---|---|---|---|
| 1 | Problem Definition & Requirement Analysis | Clarity of problem, measurable objectives, complete FR/NFR, use cases | 10 | |
| 2 | System Design | Architecture, ER/data model, form specifications, workflow design, permission matrix | 10 | |
| 3 | Forms, Reports & Pages | Variety of field types, lookups/subforms, report types, UI Builder dashboards | 15 | |
| 4 | Deluge Scripting & Workflows | Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows | 15 | |
| 5 | Integration, RBAC & AI | Zoho/external integrations, connections, roles & permissions, Zia/analytics | 10 | |
| 6 | Testing, Deployment, Security & Governance | Test coverage, environments, optimisation, security controls, audit | 10 | |
| 7 | Live Demonstration | End-to-end working of the application across roles; feature checklist verified | 15 | |
| 8 | Project Report | Adherence to template and format, clarity, figures, references | 5 | |
| 9 | Viva-Voce | Understanding of the platform, design decisions and low-code concepts | 10 | |
| | TOTAL | | 100 | |

**Table E.2:** Performance Level Descriptors

| Level | % of Max. Marks | Descriptor |
|---|---|---|
| Excellent | 90–100 | Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently. |

| Level | % of Max. Marks | Descriptor |
|---|---|---|
| Good | 70–89 | Feature implemented correctly with minor gaps; explanation adequate. |
| Satisfactory | 50–69 | Feature partially implemented or works only for the main path; limited explanation. |
| Needs Improvement | < 50 | Feature missing, non-functional or not understood by the student. |

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

## APPENDIX F – VIVA-VOCE RECORD SHEET

*(For Examiner Use Only)*

| Student Name | A VISHAL | Register No. | 241501247 |
|---|---|---|---|
| Department / Section | Artificial Intelligence And Machine Learning | Date of Viva | To be completed |

**Table F.1:** Viva-Voce Questions and Responses

| Q.No. | Question / Topic Asked | Unit | Response (E / G / S / NI) | Remarks |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |

*E – Excellent G – Good S – Satisfactory NI – Needs Improvement*

**Table F.2:** Marks Summary

| Component | Administrator | Marks Awarded |
|---|---|---|
| Criteria 1–8 (Appendix E) | 90 | |
| Viva-Voce | 10 | |
| TOTAL | 100 | |

**Examiners' Remarks:**

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____